

Topological Invariants of gapped states and locality

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based on work with [Nikita Sopenko \(IAS, Princeton\)](#), J. Math. Phys. 63 (2022) 091903, and work in progress with [Adam Artymowicz \(Caltech\)](#) and [Bowen Yang \(CMSA, Harvard\)](#).

Basics, I

In this talk d denotes the dimension of space (so space-time has dimension $d + 1$).

- **Hall conductance** σ_H is the skew-symmetric part of the conductivity tensor σ_{ij} which determines the electric current $\mathbf{J}$ as a function of the electric field $\mathbf{E}$:

$$\mathbf{J} = \sigma \mathbf{E}.$$

- At low temperatures, σ_H of insulating 2d systems becomes insensitive to variations of magnetic field and other parameters. In fact, $2\pi\sigma_H$ appears to be rational number.
- **If TQFT applies**, then the leading term in the effective action describing the response is the Chern-Simons action

$$S_{CS}^{(2)} = \frac{k}{4\pi} \int_{\mathbb{R}^3} A dA, \quad A \in \Omega^1(\mathbb{R}^3),$$

where $k = 2\pi\sigma_H$. "Quantization" of k leads to quantization of σ_H .

- Analogous terms in the effective action exist for any $d = 2n$:

$$S_{CS}^{(2n)} = \frac{k}{(2\pi)^n} \int_{\mathbb{R}^{2n+1}} A(dA)^n, \quad A \in \Omega^1(\mathbb{R}^{2n+1}).$$

For $n > 1$ they describe nonlinear response of a system to external electric and magnetic fields. The coefficient k is believed to be a rational number.

- There are also non-abelian generalizations where $U(1)$ is replaced with a compact semi-simple Lie group G . Instead of $k \in \mathbb{R}$, one has an invariant symmetric polynomial on the Lie algebra of G .
"Topological" response to external gauge fields can again be described by the Chern-Simons action.

Some questions

The above was good enough explanation for a physicist, but not for a mathematician.

- Why is it OK to use TQFT to describe response?
- Why is $k = 2\pi\sigma_H$ robust under variations of parameters?
- Is k rational?

New developments

- Zero-temperature Hall conductance and its higher-dimensional and non-abelian analogs are examples of **homotopy invariants** of ground states of "insulating" 2d systems
- To give a rigorous definition, it is crucial to work in infinite volume, hence need to use C^* -algebraic techniques
- A different interpretation of these homotopy invariants: they are **obstructions** to promoting a global symmetry of a state to a local one
- No appeal to TQFT is needed, but it serves as a motivation

Content of the talk

The original construction of invariants in [J. Math. Phys. 63 \(2022\) 091903](#) was rather mysterious and their mathematical meaning was unclear.

Goal of the talk: explain how different elements "fit together".

- Analysis (improvements of the standard C^* -algebraic approach)
- Deformation theory
- Geometry of polyhedral subsets in $\mathbb{R}^n$
- Grothendieck topologies on categories
- Cosheaves on sites

I will focus on the last two items, because their appearance is the most surprising.

Part I: Construction of invariants

Lattice spin systems: standard setup

The "lattice" $\Lambda \subset \mathbb{R}^d$ is either $\mathbb{Z}^d$ or a more general Delone subset. To each site $j \in \Lambda$ we attach a f.d. Hilbert space V_j , let $\mathcal{A}_j = \text{End}(V_j)$.

Let $\mathcal{A}$ be the norm-completion of $\bigotimes_{j \in \Lambda} \mathcal{A}_j$. It is a C^* -algebra. Its elements are **quasilocal observables**.

A state $\omega : \mathcal{A} \rightarrow \mathbb{C}$ is a positive linear functional of unit norm. Dynamics is specified by a skew-adjoint derivation of $\mathcal{A}$:

$$a \mapsto H(a) = \sum_{j \in \Lambda} [h_j, a], \quad h_j \in \mathcal{A}_j, \quad h_j^* = -h_j.$$

H is unbounded and is only defined on a dense domain $\mathcal{D} \subset \mathcal{A}$.

ω is a **gapped ground state** if $\exists \Delta > 0$ such that

$$-i\omega(a^* H(a)) \geq \Delta(\omega(a^* a) - |\omega(a)|^2), \quad \forall a \in \mathcal{D} \subset \mathcal{A}.$$

Lattice spin systems: rigging

- $\mathcal{A}$ has a natural dense Fréchet $*$ -subalgebra $\mathcal{A}_{al}$ of well-localized observables.
- We work with derivations which preserve $\mathcal{A}_{al}$:

$$b \mapsto H(b) = \sum_{j \in \Lambda} [h_j, b],$$

where $h_j \in \mathcal{A}_{al}$ is well-localized near j and uniformly bounded. Get a Fréchet-Lie algebra of "nice" derivations $\mathfrak{D}_{al}$. They have a common dense domain $\mathcal{A}_{al}$.

All Hamiltonians and generators of symmetries ("charges") are assumed to lie in $\mathfrak{D}_{al}$.

Symmetries of systems and states

A symmetry of a system is a homomorphism α from an abstract group G to the group of $*$ -automorphisms of $\mathcal{A}$.

If G is a Lie group, one needs to require $\alpha : G \rightarrow \text{Aut}(\mathcal{A})$ to be smooth in a suitable sense. On an infinitesimal level a smooth action gives a Lie algebra homomorphism $Q : \mathfrak{g} \rightarrow \mathfrak{D}_{al}$.

A G -invariant Hamiltonian $H \in \mathfrak{D}_{al}$ satisfies $[Q, H] = 0$.

A G -invariant satisfies $\omega(Q(a)) = 0$ for all $a \in \mathcal{A}_{al}$.

Definition. $F \in \mathfrak{D}_{al}$ **does not excite ω** if $\omega(F(a)) = 0$ for all $a \in \mathcal{A}_{al}$.
Such F form a Lie subalgebra $\mathfrak{D}_{al}^\omega \subset \mathfrak{D}_{al}$.

$\mathfrak{D}_{al}^\omega$ is the Lie algebra of all infinitesimal symmetries of ω . If ω is G -invariant, then Q is a homomorphism from $\mathfrak{g}$ to $\mathfrak{D}_{al}^\omega$.

$U(1)$ symmetry

In this talk I focus on $G = U(1)$, for simplicity.

For $G = U(1)$, a symmetry is determined by a "charge" $Q \in \mathfrak{D}_{al}$ such that $\exp(2\pi Q) = \text{Id}$.

A G -invariant state ω satisfies $\omega(Q(a)) = 0$ for all $a \in \mathcal{A}_{al}$. Thus $Q \in \mathfrak{D}_{al}^\omega$.

Main object of interest: derivations which do not excite ω and are $U(1)$ -invariant. These form a Fréchet-Lie algebra $\mathfrak{D}_{al}^{\omega, U(1)}$:

$$F \in \mathfrak{D}_{al}^{\omega, U(1)} \text{ iff } [Q, F] = 0 \text{ and } \omega(F(a)) = 0 \ \forall a \in \mathcal{A}_{al}.$$

Obviously, $Q \in \mathfrak{D}_{al}^{\omega, U(1)}$.

Inner derivations

A derivation $F \in \mathfrak{D}_{aI}$ does not have a well-defined average, in general. Naively, $F = \sum_j f_j$, but typically

$$\omega \left(\sum_j f_j \right) = \infty.$$

If $b = \sum_j f_j \in \mathcal{A}_{aI}$ exists, then $F(a) = [b, a]$, i.e. F is an inner derivation.

Proposition

F is inner iff $F(c) = \|c\| \times O(r^{-\infty})$, where c is a local observable and r is a distance from $0 \in \mathbb{R}^d$ to the support of c .

Thus a derivation is inner iff it is approximately localized at 0 (or any other point of $\mathbb{R}^d$).

A derivation approximately localized at a point has a well-defined average.

Approximate localization on regions

Let ω be a gapped state.

Proposition (A. Kitaev; N. Sopenko and AK)

Suppose $\Lambda = A \cup B$. Any $F \in \mathfrak{D}_{aI}$ can be written as

$$F = F_A + F_B,$$

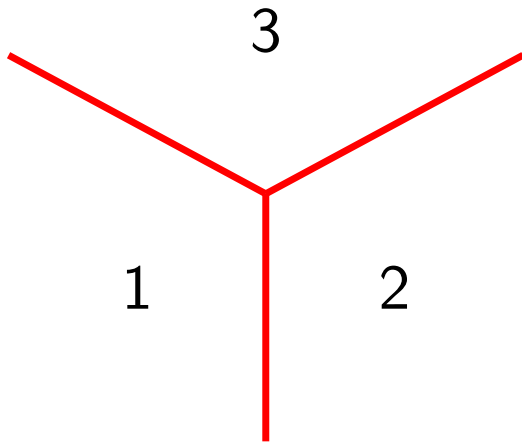
where F_A and F_B are approximately localized on A and B respectively. If $F \in \mathfrak{D}_{aI}^\omega$, one can choose F_A and F_B to be in $\mathfrak{D}_{aI}^\omega$. If in addition ω and F are $U(1)$ -invariant, one can choose F_A and F_B to be in $\mathfrak{D}_{aI}^{\omega, U(1)}$.

This is the key technical result which enables the construction of Hall conductance and other invariants of **gapped** states.

From now on, I will use a shorthand $\mathfrak{D} \equiv \mathfrak{D}_{aI}^{\omega, U(1)}$.

Hall conductance

Let $Q \in \mathfrak{D}$ and $\Lambda \subset \mathbb{R}^2$. Partition $\mathbb{R}^2$ into three sectors.



Write $Q = Q_1 + Q_2 + Q_3$ where $Q_i \in \mathfrak{D}$ is approximately localized in region i .
Let $Q_{ij} = [Q_i, Q_j]$.

Q_{12} is localized near the ray 12. But

$$[Q_1, Q_2] = [Q - Q_2 - Q_3, Q_2] = -[Q_3, Q_2],$$

so Q_{12} is also localized near the ray 23.

Therefore Q_{12} is localized near the vertex 123, so it is inner and $\sigma_{d=2}^\omega = -i\omega(Q_{12}) \in \mathbb{R}$ is well-defined.

Main theorem

Theorem (N. Sopenko and AK)

$\sigma_{d=2}^\omega$ does not depend on any choices made (except orientation of $\mathbb{R}^2$). It is also unchanged under $\omega \mapsto \omega \circ \alpha$ where α is an LGA. It is proportional to the zero-temperature Hall conductance of the state ω .

One can generalize to arbitrary Lie groups. Then $\sigma_{d=2}^\omega$ takes values in G -invariant quadratic polynomials on the Lie algebra of G .

For special classes of states (such as states of the form $\omega_0 \circ \alpha$ where ω_0 is factorized pure state and α is any LGA) $\sigma_{d=2}^\omega$ is quantized (integral with suitable normalization).

Quantization of Hall conductance was first proved by M. Hastings and S. Michalakis by a very different method.

Part II: Formal developments

Questions

- How do we generalize to $d > 2$?
- Why conical regions?
- Can one get more invariants using other configurations of regions?

Approximate locality of nice derivations

- For any $U \subset \Lambda$ can define $\mathfrak{D}(U) \subset \mathfrak{D} = \mathfrak{D}_{al}^{\omega, U(1)}$, a Lie subalgebra of derivations approximately localized on U .
- If $\Lambda = \cup_i U_i$ and $Q = \mathfrak{D}(\Lambda) = \mathfrak{D}$, can write $Q = \sum_i Q_i$, where $Q_i \in \mathfrak{D}(U_i)$.
- The decomposition $Q = \sum_i Q_i$ is unique up to derivations "living" on $U_i \wedge U_j$ which is a "generalized intersection" of U_i and U_j .

Assignment of vector spaces to regions is reminiscent of presheaves of vector spaces.

The appearance of unions and intersections of regions is reminiscent of sheaves.

Actually, a closer analogy is with **pre-cosheaves and cosheaves**.

Pre-cosheaves of vector spaces

Let X be a topological space.

Definition

A pre-cosheaf $\mathcal{F}$ of vector spaces on X is an assignment $U \mapsto \mathcal{F}(U) \in \mathbf{VECT}$ to every open $U \subset X$ and a linear map $\rho_{VU} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$ to every inclusion of opens $U \subset V$ such that $\rho_{WV} \circ \rho_{VU} = \rho_{WU}$ for all $U \subset V \subset W$.

The map ρ_{VU} is called **co-restriction**. For pre-sheaves, the map is in the opposite direction and is called restriction.

Example: take $\mathcal{F}(U)$ to be the space of continuous functions on X with support in U .

In this example ρ_{VU} is injective for all $U \subset V$. Such pre-cosheaves are called **coflasque**. Let's restrict to those, then can suppress the maps ρ_{VU} from notation.

Cosheaves of vector spaces

Definition

A (coflasque) cosheaf is a pre-cosheaf with the "Mayer-Vietoris property": $\forall U_1, U_2$ and $\forall Q \in \mathcal{F}(U_1 \cup U_2)$ one can write $Q = Q_1 + Q_2$ for some $Q_i \in \mathcal{F}(U_i)$. Q_i are unambiguous up to

$$Q_1 \mapsto Q_1 + P_{12}, \quad Q_2 \mapsto Q_2 - P_{12}, \quad P_{12} \in \mathcal{F}(U_1 \cap U_2).$$

This can be expressed as the exactness of the sequence

$$0 \leftarrow \mathcal{F}(U_1 \cup U_2) \leftarrow \mathcal{F}(U_1) \oplus \mathcal{F}(U_2) \leftarrow \mathcal{F}(U_1 \cap U_2)$$

Actually, in the coflasque case we get a short exact sequence

$$0 \leftarrow \mathcal{F}(U_1 \cup U_2) \leftarrow \mathcal{F}(U_1) \oplus \mathcal{F}(U_2) \leftarrow \mathcal{F}(U_1 \cap U_2) \leftarrow 0$$

Sheaves and cosheaves on sites

To use these notions in our case, we need to deal with two issues:

- Λ is a very boring (discrete) topological space
- Localization in our case is only approximate, so U_i are "fuzzy". Then what do we mean by intersection, union, and cover?

M. Kashiwara and P. Schapira (2001): one can organize functions with growth conditions into a sheaf if we replace the category $Open(X)$ with a different category equipped with a "Grothendieck topology". A category equipped with a Grothendieck topology is called a **site**.

Similarly, functions with decay conditions (test functions) can be organized into a cosheaf on a suitable site.

Semilinear sets

Definition

A semilinear set is a subset of $\mathbb{R}^d$ defined by a finite system of linear equations and inequalities.

A **closed semilinear set** is the same as a finite union of polyhedra. A polyhedron is the same as a closed convex semilinear set.

Let's use the metric $d(x, y) = \|x - y\|_\infty$.

Proposition

For any $U \subset \mathbb{R}^d$ and $r \geq 0$ let U^r be the r -thickening of U . If U is semilinear (polyhedron), then U^r is semilinear (polyhedron).

Definition

Let CS_d be the set of non-empty closed semilinear sets with a pre-order:
 $U \leq V$ iff $U \subseteq V^r$ for some $r \geq 0$.

A category of closed semilinear sets

- A pre-ordered set can be thought of as a category: $Mor(U, V)$ is a singleton if $U \leq V$ and empty otherwise.
- A pre-ordered set is like a poset, except $U \leq V$ and $V \leq U$ does not imply $U = V$. But it implies U and V are isomorphic objects.
- In CS_d , **all non-empty bounded U are isomorphic**. But cones with the same vertex and different bases are not isomorphic.
- In fact, CS_d is equivalent to the category of closed polyhedral subsets of S^{d-1} (sphere at infinity).
- An **initial object** in CS_d is any bounded closed semilinear subset (for example, $0 \in \mathbb{R}^d$).

Coherent topology on CS_d

- CS_d has all finite coproducts (joins). One can take the join $U \vee V$ to be $U \cup V$.
- CS_d has all finite products (meets, generalized intersections). For polyhedra, one can take the meet $U \wedge V$ to be $U^{d(U,V)} \cap V^{d(U,V)}$, where $d(U, V)$ is the distance between U and V .

Products and coproducts are unique up to isomorphism.

Definition

A cover of U is a finite nonempty collection $\{U_i\}_{i \in I}$ such that $U_i \leq U$ $\forall i \in I$ and $U \leq \bigvee_{i \in I} U_i$.

This makes CS_d into a site.

Spherical polyhedra

Definition

A polyhedral subset of S^{d-1} is a finite union of spherical polyhedra.

Theorem

The category CS_d is equivalent to the category of polyhedral subsets of S^{d-1} (with inclusions as morphisms).

This is because every polyhedral subset of $\mathbb{R}^d$ is isomorphic to a polyhedral cone with apex at 0.

Thus we can view sheaves and cosheaves on the site CS_d as sheaves and cosheaves on the sphere "at infinity".

A subtlety

The empty subset of S^{d-1} corresponds to the isomorphism class of a bounded polyhedron in $\mathbb{R}^d$ (for example, the point 0). The latter is an initial object in the category of closed semilinear sets.

Normally, one allows empty covers of the empty set.

Here it is more natural not to allow empty covers, because in $\mathbb{R}^d$ the initial object is not "empty".

The cosheaf $\mathfrak{D}$ and its Čech complex

Proposition

The assignment $U \mapsto \mathfrak{D}(U)$ is a coflasque cosheaf of vector spaces on the site CS_d .

Note that for U **bounded** semilinear (the analog of empty set), $\mathfrak{D}(U) \neq 0$. This is because we do not allow the empty cover of the initial object.

For any cover $\mathfrak{U} = \{U_i\}_{i \in I}$ of $\mathbb{R}^d$, and any pre-cosheaf $\mathcal{F}$ one can define the homological Čech complex $C_\bullet(\mathfrak{U}, \mathcal{F})$:

$$0 \leftarrow \mathcal{F}(\Lambda) \leftarrow \bigoplus_i \mathcal{F}(U_i) \leftarrow \bigoplus_{i < j} \mathcal{F}(U_i \wedge U_j) \leftarrow \bigoplus_{i < j < k} \mathcal{F}(U_i \wedge U_j \wedge U_k) \leftarrow \dots$$

For a coflasque cosheaf, the homology of this complex is trivial.

Lie algebra structure

$\mathfrak{D}$ is a pre-cosheaf of Lie algebras, i.e. for any $U \leq V$ the inclusion $\mathfrak{D}(U) \rightarrow \mathfrak{D}(V)$ is a Lie algebra homomorphism.

$\mathfrak{D}$ is not a cosheaf of Lie algebras. But we have the following "**Property I**": $\forall U, U'$, we have $[\mathfrak{D}(U), \mathfrak{D}(U')] \subset \mathfrak{D}(U \wedge U')$. This property encodes locality.

Equivalently, the co-restriction maps $\mathfrak{D}_{al}(U) \rightarrow \mathfrak{D}_{al}(V)$ are inclusions of Lie ideals. We call such a structure a "**local Lie algebra**" over a site.

Definition

A local Lie algebra over a site is a coflasque pre-cosheaf of Lie algebras which satisfies Property I and is a cosheaf of vector spaces.

Next step: define a functor from the category of local Lie algebras to a **Differential Graded Lie Algebra (DGLA)**.

Reminder on DGLAs

A DGLA is a chain complex of vector spaces $(\mathcal{L}, \partial)$ equipped with a bilinear operation $[\cdot, \cdot]$ of degree 0 such that

- $[a, b] = -(-1)^{|a|\cdot|b|}[b, a]$
- $[\cdot, \cdot]$ obeys a graded version of the Jacobi identity
- $\partial[a, b] = [\partial a, b] + (-1)^{|a|}[a, \partial b]$ (super-Leibniz)

A DGLA $\mathcal{L}$ is called **acyclic** if its homology (as a chain complex) is trivial.

Let $[\mathcal{L}, \mathcal{L}]$ be the vector space whose elements are finite sums of commutators of elements of $\mathcal{L}$. It is also a DGLA (the commutator DGLA). $[\mathcal{L}, \mathcal{L}]$ need not be acyclic even if $\mathcal{L}$ is acyclic.

DGLA attached to a cover

Let $\mathfrak{U} = \{U_i\}_{i \in I}$ be a cover of a site. Let $\mathcal{F}$ be a local Lie algebra over the same site.

Proposition

The Čech complex $C_{\bullet+1}(\mathfrak{U}, \mathcal{F})$ is a Differential Graded Lie Algebra.

The DGLA grading is shifted w.r. to the standard Čech grading so that $\bigoplus_i \mathcal{F}(U_i)$ is in degree 1. I will use the standard Čech grading, then $[\cdot, \cdot]$ has degree +1.

The graded Lie bracket is the skew-symmetrization of the usual commutator. For example, let $a = \bigoplus_i a_i$ and $b = \bigoplus_i b_i$ be two 0-chains. Their bracket is a 1-chain with components

$$[a_i, b_j] - [a_j, b_i] \in \bigoplus_{i < j} \mathcal{F}(U_i \wedge U_j).$$

The case $d = 2$: Hall conductance

We have $Q \in \mathfrak{D}(\mathbb{R}^2)$ which is central: $[Q, \cdot] = 0$. Pick a cover $\mathfrak{U}$ of $\mathbb{R}^2$. There exists a 0-chain $q = \bigoplus_{i \in I} q_i$ such that $Q = \sum_i q_i$.

Consider a 1-chain $p = \frac{1}{2}[q, q]$. **It is a cycle.**

$\mathfrak{U}$ induces a cover $\tilde{\mathfrak{U}}$ of S^1 by closed intervals. Let α be a 1-cochain of the Čech complex $C^\bullet(\tilde{\mathfrak{U}}, \mathbb{R})$. Let $p(\alpha) = \sum_{i < j} \alpha_{ij} p_{ij} \in \mathfrak{D}(\Lambda)$.

Theorem

If α is a cocycle, then $p(\alpha)$ is inner, and thus $\omega(p(\alpha)) \in \sqrt{-1}\mathbb{R}$ is well-defined. It does not depend on the choice of q . It depends on $\mathfrak{U}$ and α only through the class of α in $\varinjlim \check{H}^1(\tilde{\mathfrak{U}}, S^1) \simeq H^1(S^1)$.

Since $H^1(S^1)$ is one-dimensional, the Hall conductance invariant is the unique invariant one can construct from the 1-cycle $[q, q]$.

Hall conductance as an obstruction to gauging

The Lie algebra of $U(1)$ is $\mathbb{R}$. For any polyhedral U let $\mathbb{R}_{aI}(U)$ be the space of bounded functions $\Lambda \rightarrow \mathbb{R}$ which decays superpolynomially away from U . $U \mapsto \mathbb{R}_{aI}(U)$ is a **local Lie algebra**. Let's call it $\mathbb{R}_{aI}$.

$\mathbb{R}_{aI}$ encodes $U(1)$ gauge symmetry on a lattice. Global symmetry corresponds to the image of the obvious homomorphism $\mathbb{R} \rightarrow \mathbb{R}_{aI}(\mathbb{R}^n)$.

A global $U(1)$ symmetry of a gapped state ω gives a Lie algebra morphism $\mathbb{R} \rightarrow \mathfrak{D}_{aI}^\omega(\mathbb{R}^n)$. What does it mean to gauge this $U(1)$ symmetry?

It is a lift of the Lie algebra morphism $\mathbb{R} \rightarrow \mathfrak{D}_{aI}^\omega(\mathbb{R}^n)$ to a morphism of local Lie algebras $\mathbb{R}_{aI} \rightarrow \mathfrak{D}_{aI}^\omega$.

Theorem

If the Hall conductance of a 2d state ω is nonzero, such a lift does not exist.

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